CISCO Certified Network Associate (CCNA)

Skill Final Answer – Semester 01

Roll: 15

Name: Pritom Bhowmik

(Batch 13)

Problem Statement: Suppose you are working a company as a network engineer. You have been given an 10.10.0.0/18 . You must provide for the following subnet arrangement for the company:

(a) Subnet_1 up to 56 hosts
(b) Subnet_2 up to 32 hosts
(c) Subnet_3 up to 50 hosts
(d) Point to point link between Subnet_1 and Subnet_2 2 hosts
(e) Point to point link between Subnet_1 and Subnet_3 2 hosts
(f) Point to point link between Subnet_2 and Subnet_3 2 hosts

Fill up the following table and test the connectivity among all of the PCs through Packet Tracer implementation.

Subnet Name	Needed Size	Allocated Size	Network Address	Mask	Assignable Range	Broadcast Address
Subnet_1						
Subnet_2						
Subnet_3						
Between Subnet_1 & Subnet_2						
Between Subnet_1 & Subnet_3						
Between Subnet_2 & Subnet_3						

Solution of this problem:

IP = 10.10.0.0/18

18 = 11111111.11111111.11000000.00000000

At first Sort (Descending order) the number of subnets according to their requirement

(a) Subnet_1 = 56 +2 = 58

(b) Subnet_2 = 32+2 = 34

(c) Subnet_3 = 50+2 = 52

(d) Point to point link between Subnet_1 and Subnet_2 = 2+2 = 4

(e) Point to point link between Subnet_1 and Subnet_3 = 2+2 = 4

(f) Point to point link between Subnet_2 and Subnet_3 = 2+2 = 4

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1		
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Network Subnet_1	10.10.0.0/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1st Host of Subnet_1	10.10.0.1/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	Last Host of Subnet_1	10.10.0.62/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	Broadcast Subnet_1	10.10.0.63/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	Network Subnet_2	10.10.0.64/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1st Host of Subnet_2	10.10.0.65/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	Last Host of Subnet_2	10.10.0.126/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	Broadcast Subnet_2	10.10.0.127/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	Network Subnet_3	10.10.0.128/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1st Host of Subnet_3	10.10.0.129/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	0	Last Host of Subnet_3	10.10.0.190/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	Broadcast Subnet_3	10.10.0.191/26
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	Network Subnet_1 to 2	10.10.0.192/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	1st Host Subnet_1 to 2	10.10.0.193/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	1	0	Last Host Subnet_1 to 2	10.10.0.194/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1	Broadcast Subnet_1 to 2	10.10.0.195/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	0	0	Network Subnet_1 to 3	10.10.0.196/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	0	1	1st Host Subnet_1 to 3	10.10.0.197/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	1	0	Last Host Subnet_1 to 3	10.10.0.198/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	1	1	Broadcast Subnet_1 to 3	10.10.0.199/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0	0	Network Subnet_2 to 3	10.10.0.200/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0	1	1st Host Subnet_2 to 3	10.10.0.201/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	Last Host Subnet_2 to 3	10.10.0.202/30
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1	1	Broadcast Subnet_2 to 3	10.10.0.203/30

Subnet Name	Needed Size	Allocated Size	Network Address	Mask	Assignable Range	Broadcast Address
Subnet_1	56	62	10.10.0.0/26	255.255.255.192	10.10.0.1 - 10.10.0.62	10.10.0.63
Subnet_2	32	62	10.10.0.64/26	255.255.255.192	10.10.0.65 - 10.10.0.126	10.10.0.127
Subnet_3	50	62	10.10.0.128/26	255.255.255.192	10.10.0.129 - 10.10.0.190	10.10.0.191
Between Subnet_1 & Subnet_2	2	2	10.10.0.192/30	255.255.255.252	10.10.0.193 - 10.10.0.194	10.10.0.195
Between Subnet_1 & Subnet_3	2	2	10.10.0.196/30	255.255.255.252	10.10.0.197 - 10.10.0.198	10.10.0.199
Between Subnet_2 & Subnet_3	2	2	10.10.0.200/30	255.255.255.252	10.10.0.201 - 10.10.0.202	10.10.0.203